Knowledge-Based Systems 284 (2024) 111323

Contents lists available at ScienceDirect

Knowledge-Based Systems

journal homepage: www.elsevier.com/locate/knosys

LMKG: A large-scale and multi-source medical knowledge graph for intelligent medicine applications

Peiru Yang [a,1], Hongjun Wang [a,1], Yingzhuo Huang [a], Shuai Yang [c], Ya Zhang [c], Liang Huang [c], Yuesong Zhang [c], Guoxin Wang [d,c], Shizhong Yang [e,f], Liang He [a,b,*], Yongfeng Huang [a,f]

[a] Department of Electronic Engineering, Tsinghua University, Beijing 100084, China
[b] Beijing National Research Center for Information Science and Technology, Tsinghua University, Beijing 100084, China
[c] JD Health International Inc., Beijing 101111, China
[d] College of Biomedical Engineering and Instrument Science, Zhejiang University, Hangzhou 310027, China
[e] Hepatopancreatobiliary Center, Beijing Tsinghua Changgung Hospital, Beijing 102218, China
[f] Institute for Precision Medicine, Tsinghua University, Beijing 102218, China

ARTICLE INFO

Keywords:
Medical knowledge graph
Intelligent medicine
Knowledge engineering

ABSTRACT

Medical Knowledge Graph (KG) has shown great potential in various healthcare scenarios, such as drug recommendation and clinical decision support system. The factors that determine the role of a medical KG in practical applications are the scale, coverage, and quality of the medical knowledge it can provide. Most existing medical KGs are extracted from a single or a few information sources. However, medical knowledge extracted from insufficient information sources is usually highly incomplete or even biased, which results in a lack of data completeness and may lessen their effectiveness in real-world scenarios. Besides, the coverage of entity and relation types is inadequate in most previous works, which also might restrict their potential usage in future applications. In this paper, we build a unified system that can extract and manage medical knowledge from heterogeneous information sources. We first employ named entity recognition and relation extraction methods to extract knowledge triplets from medical texts. Then we propose a hierarchical entity alignment framework for further knowledge refinement. Based on our system, we construct a large-scale, high-quality, multi-source, and multi-lingual medical KG named *LMKG*, which includes 13 entity types and 17 relation types, and contains 403,784 entity and 1,225,097 relation instances. We conduct extensive experiments to evaluate the quality of *LMKG*. Experimental results show that *LMKG* can effectively enhance the performance of both upstream and downstream intelligent medicine applications. We have publicly released the KG resources and corresponding management service interface to facilitate research and applications in the medical field.

1. Introduction

Knowledge graph (KG) is an important method in intelligent systems to summarize expert knowledge from big data, in order to provide prior knowledge for artificial intelligence (AI) algorithms. With the development of machine learning (ML) and deep learning (DL), data-driven medical AI algorithms now attract more attention in both industry and academia. In general, it requires massive expert experience to provide high-quality medical services. However, due to the constraints of data availability and the deficiencies of data management, training data is usually insufficient in real-world medicine tasks [1,2], making it difficult for many intelligent algorithms to summarize adequate expert knowledge for decision-making. Fortunately, KGs extracted from medicine data usually contain abundant expertise, which can be used to guide the decision of various intelligent medicine algorithms and enhance their performance. Thus, medical KG plays an important role in various applications including drug discovery [3–5], clinical decision support system [6–8], and medicine recommendation [9,10]. In general, the construction and application of medical KGs have drawn increasing attention in recent years [11].

In a medical KG, entities generally represent medical concepts, and the relations between them reflect their connections in the medical domain. For instance, Fig. 1 gives a toy example of the structure of medical KGs, showing the relations between the disease concept "Gallstone" and some other medical concepts. In recent years some

* Corresponding author at: Department of Electronic Engineering, Tsinghua University, Beijing 100084, China.
E-mail addresses: ypr21@mails.tsinghua.edu.cn (P. Yang), hj-wang21@mails.tsinghua.edu.cn (H. Wang), huangyingzhuo17@gmail.com (Y. Huang), yangshuai15@jd.com (S. Yang), ysza02008@btch.edu.cn (S. Yang), heliang@mail.tsinghua.edu.cn (L. He).

[1] Equal contributions.

https://doi.org/10.1016/j.knosys.2023.111323
Received 21 April 2023; Received in revised form 20 November 2023; Accepted 20 December 2023
Available online 23 December 2023

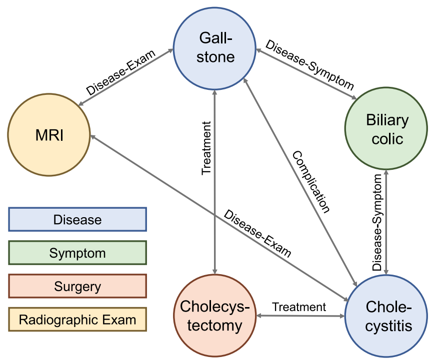

Fig. 1. A toy example of the entities and relations in the KG.

works have focused on extracting knowledge triplets and built KGs for the medical domain [12–14]. For example, Li et al. [14] extract knowledge from a large number of electrical medical records (EMRs) in a single hospital and build a KG containing 22,508 entities and 579,094 relations. In general, many existing medical KGs are extracted from a single information source. However, the knowledge provided by a single-source medical KG is usually highly incomplete or even biased, which might be ineffective for real-world tasks. To tackle this challenge, a few works constructed medical KG from multiple information sources. For example, Cheng et al. [15] construct a stroke knowledge graph based on several sources including vertical medical websites, crowd-sourced websites, and a Chinese symptom library. In general, medical knowledge derived from different sources varies in quality and reliability. Therefore, knowledge source information is informative for indicating the quality of a medical KG and therefore is important for usage in real-world scenarios. However, the knowledge source information is missing in most existing medical KGs, limiting their effectiveness in various downstream applications. In addition, many previous KGs only contain a few entity and relation types. It is difficult for a medical KG with insufficient knowledge types to provide sufficient information to guide the decisions of complex real-world medical tasks, possibly affecting the KG's effectiveness in downstream tasks. Thus, it is still important to build a large-scale, multi-source, and high-quality medical KG.

Therefore, we propose a unified system for the extraction of medical KG from heterogeneous medical texts. First, we collect data from various sources such as medical wikis, medical journals, and national official websites on medicine, and then process them to form a unified text structure. Then we use a pre-trained named entity recognition (NER) model to extract entities and a pre-trained relation extraction (RE) model to identify the relations between entities. Furthermore, we propose a hierarchical entity alignment method to refine the knowledge triplets extracted from different sources. Based on the system, we build a large-scale, multi-source, and high-quality medical KG (named *LMKG*) containing 403,784 entities and 1,225,097 relations, classified into 13 entity types and 17 relation types. Knowledge source information is also retained in the KG to preserve the variability of different information sources. We also generate graph embeddings of the KG to facilitate its application in downstream tasks. We conduct extensive experiments to evaluate the qualities of different KGs. Experimental results show that *LMKG* can consistently outperform other public medical KGs in improving the performance of downstream tasks such as medical NER and medical RE. We have released *LMKG* to the public, along with a Neo4j-based management service interface, in the hope of assisting medical research and applications. Detailed instructions on KG data and system resources can be found on our page: https://github.com/ypr17/LMKG.

2. Related work

2.1. Knowledge graph construction

Knowledge graph is a widely used method for real-world knowledge representation and has become the foundation of many intelligent information systems [11]. Most KGs transform information in the real world into structured data and are composed of knowledge triplets, which are in the form of "*head entity - relation - tail entity*". Therefore, the key to KG construction lies in the extraction of knowledge triplets from unstructured textual data (e.g., online wiki websites). In the commonly used KG construction paradigm, a NER model is usually used to extract the entities from the raw texts, and a RE model is used for the identification of the connections between a pair of entities. Next, we will introduce the previous work in these two areas.

2.1.1. Named entity recognition

In recent years, NER algorithms have been widely studied in many previous works [16–18]. State-of-the-art NER methods are usually based on deep learning models [19]. In DL-based methods, the NER task is formulated as an end-to-end sequence labeling problem. The model accepts original sentences as input, and each token is transformed into vectors via the embedding technique. Then the DL-based methods usually capture the contexts from the embedding vector sequences via various neural models such as Long Short-Term Memory (LSTM) [20] and Gate Recurrent Unit (GRU) [21], and finally decode the label sequence from them. For example, Huang et al. [16] propose the commonly used Bi-LSTM-CRF model, in which the LSTM uses both past and future input features, and the CRF layer detects label transition dependencies. Following Huang et al. [16], Chiu and Nichols [17] use a Bi-LSTM-CNN model incorporating a bidirectional LSTM and a character-level CNN. Some word-level features such as capitalization and partial lexicon matches are also encoded in the model. Yang et al. [18] employ deep GRU on both character and word-level encoding. The model is also extended to multi-task and cross-lingual joint training. However, most existing NER methods are based on large-scale text corpora, while the valid data is usually insufficient for model training in the medical domain. Moreover, the presence of a large amount of terminology and expertise in medical texts makes the task more challenging for regular NER models.

Therefore, different approaches are applied to adapt DL-based NER models to medical texts [22–24]. Some existing works propose to utilize dictionary knowledge to guide NER models. For example, Wang et al. [22] incorporate medical dictionaries into data-driven DL models using different embedding strategies. Li et al. [23] propose an RNN-based approach for the Chinese Bio-NER problem combining word embeddings and character embeddings to capture orthographic and lexicon semantic features. Some other existing works explore improving the performance of medical NER based on the pre-training technique. For example, Giorgi and Bader [25] transfer a DNN trained on a large, noisy corpus to a smaller but more reliable one for better performance, in order to solve the problem of insufficient data in biomedical NER. Dai et al. [24] compare a variety of models for Chinese medical NER, and found that the BERT-BiLSTM-CRF model outperforms other models for the task. In our work, we implement the NER model for KG construction based on [24] since it has the best empirical performance in our experiments.

2.1.2. Relation extraction

As a crucial task for semantic analysis, RE aims to automatically extract relationships between co-occurring entities. Similar to NER, RE methods are now mainly dominated by DL models [26,27]. Nguyen and Grishman [28] propose an RE model consisting word and position embedding layer, CNN layer, and max-pooling layer, with multiple window sizes for convolutional kernel. Miwa and Bansal [26] provide an end-to-end model that stacks bidirectional tree-structured Bi-LSTM

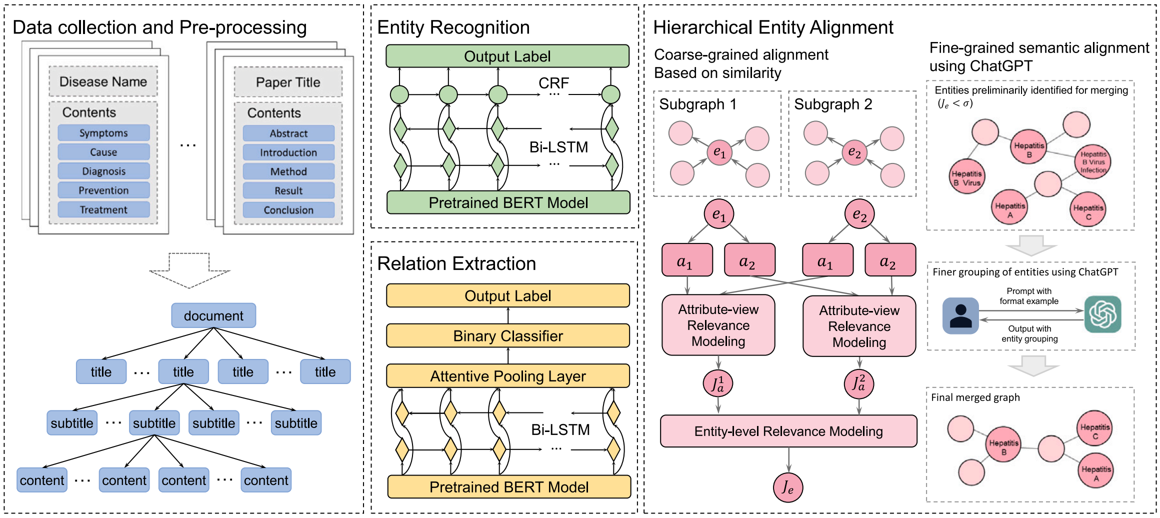

Fig. 2. The framework of the proposed procedure for medical KG extraction from multiple information sources.

over sequential Bi-LSTM, to capture both word sequence and dependency tree substructure information in the text. Guo et al. [27] propose to use the soft-pruning strategy to selectively employ the information provided by dependency trees via an attention-guided layer. However, it may be sub-optimal to apply these general RE methods to the medical domain, since RE for medical corpora is a more demanding task considering the specialization of the medical field.

In the medical scenarios, researchers have employed a variety of techniques to improve the performance of RE models [29–31]. For example, Zhang et al. [29] propose an attention-based model to extract the multi-aspect semantic information for the Chinese medical relation extraction. A multi-hop attention mechanism is applied for semantic representation. Ruan et al. [30] extract multiple types of relations from the Chinese medical records with a novel heterogeneous graph representation learning method. Zhang et al. [32] propose an attention-based deep residual network model to recognize medical concept relations in Chinese medical records. Qi et al. [31] propose a knowledge-enhanced BERT-CNN-LSTM medical RE framework utilizing medical knowledge from a medicine dictionary. In our work, we implement the RE model for KG extraction based on Qi et al. [31] with some adaptions.

2.2. Medical knowledge graph

Since the abundant expertise contained in medical KGs is highly helpful to intelligent medicine algorithms, some researchers have constructed medical KGs based on medical data [12–14].

Some early works construct medical KGs on a small scale, and the entity and relation types are relatively limited. For example, Finlayson et al. [33] construct a set of co-occurrence matrices from clinical notes, quantifying the relatedness between medical concepts including drugs, diseases, procedures, and devices. Zhou et al. [34] generate a symptom-based network of human diseases including 4219 diseases based on large-scale medical bibliographic records. In general, these early works are not effective enough in real-world applications limited by scale and construction methods.

More recent works have constructed medical KGs with larger scales, more entity and relation types, and better extraction methods. Among them, most KGs are extracted from a single information source. For example, Rotmensch et al. [12] build a disease–symptom graph from 273,174 patient records using rudimentary concept extraction. Zhao et al. [13] extract medical entities from EMRs to form a medical knowledge network (EMKN) containing 6733 nodes and 154,462 edges. However, only two entity types (disease and symptom) are included, which can hardly capture the complexity of medical knowledge of the real world. Li et al. [14] propose a systematic procedure for KG construction and a quadruplet structure to represent medical knowledge. A medical KG with 22,508 entities and 579,094 quadruplets is established based on single-source EMRs, with downstream experiments proving its efficiency. Weng et al. [35] construct a traditional Chinese medicine KG from treatment data records, including 59,882 entities, 17 relations, and 604,700 triples.

There are also a few works providing multi-source medical knowledge in their KGs. For example, Cheng et al. [15] build a knowledge graph targeted at the disease of stroke containing 4113 entities and 9230 relations. It is extracted from multiple information sources, including vertical medical websites, crowd-sourced websites, and existing knowledge bases. Zan et al. [36] construct a Chinese symptom knowledge base (CSKB) containing 8722 symptom entities, along with 146 631 relations between the symptoms and related diseases, body parts stroke, and suffering population. Ernst et al. [37] construct a medical knowledge graph named Knowlife from different Web sources, including scientific publications, health portals, and online communities.

At the same time, many works focus on how to embed medical knowledge into vector representations. Celebi et al. [38] use different methods to embed knowledge graphs for predicting unknown drug–drug interactions. Li et al. [39] propose an embedding algorithm for medical KGs that can learn the probability values of triplets into representation vectors. They evaluate the proposed embedding method using a link prediction task. Gong et al. [40] bridge EMRs and drug databases together to construct a heterogeneous graph containing diseases, drugs, and patients. Then, the entities are simultaneously embedded and used for drug recommendation based on patients' medical diagnoses. Generally, the techniques for knowledge representation learning and the construction of the knowledge graph itself have advanced in tandem, mutually enhancing one another.

Note that most of these works on medical KG construction do not provide open-source resources of their work to the public, and this lack of accessibility may restrict the potential applications of their KGs. Also, existing works often overlook the inclusion of knowledge source information, which might be crucial for ensuring knowledge accuracy and reliability. Moreover, the limited number of knowledge types and knowledge triplets in existing works hinders their effectiveness in providing comprehensive and diverse information for various downstream medical tasks.

Different from these works mentioned above, *LMKG* is a large-scale, multi-source, and high-quality medical KG containing 403,784 entity instances and 1,225,097 relation instances, classified into 13 entity types and 17 relation types. (We also provided a detailed comparison between different KGs in Table 1.)

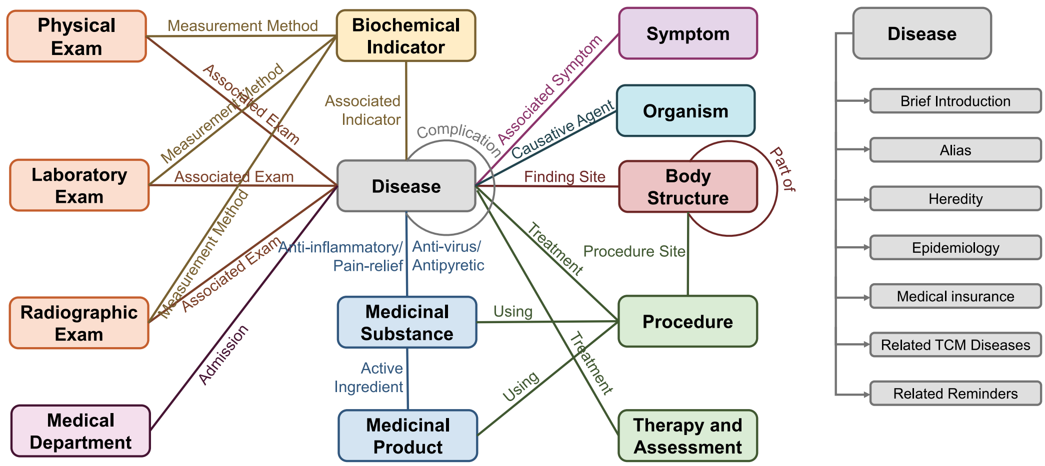

Fig. 3. The metadata structure for entity and relation types, as well as attributes.

Table 1
The comparison between *LMKG* and other medical KGs in scale, number of entity and relation types, and availability of resources.

KG	# entities	# relations	# entity types	# relation types	Multi-source	Open-source
Zhao et al. [13]	6733	154,462	5	1	×	×
Rotmensch et al. [12]	647	3120	2	1	×	✓
Shen et al. [41]	289	618	3	1	×	×
Li et al. [14]	22,508	579,094	9	7	×	×
Zan et al. [36]	8772	146,631	3	8	✓	✓
Cheng et al. [15]	4113	9230	6	10	✓	×
LMKG	403,784	1,225,097	13	17	✓	✓

3. System

In this section, we will introduce the construction approach for our large-scale, multi-source and high-quality medical KG named *LMKG*. We will first elaborate on the entity and relation type definitions in *LMKG*, and then the multi-source KG extraction and management system. The extraction procedure is shown in Fig. 2, which can be divided into four steps: data collection and pre-processing, entity recognition, relation extraction, and hierarchical entity alignment.

3.1. Entity and relation type definition

Fig. 3 presents the metadata structure of *LMKG*. It demonstrates all the entity types, relation types, and entity attributes defined in *LMKG*. Specifically, we have defined entity and relation types based on Snomed CT, a widely used clinical terminology in the medical field [42]. Snomed CT provides a comprehensive set of standardized concepts that enable precise representation of various clinical entities and their relationships. By leveraging Snomed CT, we ensure that the defined entity and relation types align with established medical terminologies and practices. To meet the requirements of real-world intelligent medicine applications, we have also sought advice from domain experts to make certain modifications and adjustments to the entity and relation types. This collaborative approach ensures that *LMKG* effectively addresses the specific requirements of medical applications, ultimately improving its accuracy and usefulness in clinical settings. There are 13 entity types and 17 relation types in total, and the detailed entity and relation type definitions are summarized as follows.

3.1.1. Entity type definition

Disease: As the backbone of *LMKG*, *Disease* entities are directly extracted from the information sources and linked to other entity types. Most of the *Disease* entities are equipped with multiple attributes, such as alias, heredity, epidemiology, related traditional Chinese medicine (TCM) diseases, etc.

Symptom: Each *Symptom* entity is linked to a *Disease* entity to describe its clinical symptoms.

Procedure: The names of medical and surgical procedures are regarded as *Procedure* entity names.

Body Structure: The names of body parts as well as anatomical terminologies of the human body are collected as *Body Structure* entities.

Medicinal Substance: The substance of drugs and healthcare products for both Western and Chinese medicine are classified into *Medicinal Substance* entities. Note that most of the *Medicinal Substance* entities are not formal medicine names, but rather common names for drugs, ingredients, or source materials.

Medicinal Product: The formal names of medicines regulated by officials (e.g. National Medical Products Administration) are used as *Medicinal Product* entities. They are related to *Medicinal Substance* entities to demonstrate the correspondence between the formal product name and the ingredients or common name.

Organism: Organisms such as bacteria, viruses, fungi, microorganisms, and parasites are collectively referred to as *Organism* entities.

Biochemical Indicator: The indicators of the liver, kidney, blood glucose, blood lipids, protein content, certain enzymes, etc. are known clinically as biochemical indicators. Therefore, the names of these tests are regarded as *Biochemical Indicator* entities.

Laboratory Exam: Laboratory tests refer to the examination of the blood sample, urine sample, or body tissues, and are collected into *Laboratory Exam* entities.

Radiographic Exam: The radiographic examinations such as CT, X-ray, MRI, etc., are collected into the category of *Radiographic Exam*.

Physical Exam: Physical examinations refer to the detection and measurement of the level of morphological structure and functional development for the human body and are collected into *Physical Exam* entities.

Medical Department: Names of different medical departments are used as *Medical Department* entities.

Therapy and Assessment: The *Therapy and Assessment* category includes therapies, assessment scales, and other medical treatments aimed at assessing and improving health status.

3.1.2. Relation type definition

Associated Symptom: If a *Symptom* entity is the corresponding symptom of a *Disease* entity, an *Associated Symptom* relation exists between them.

Finding Site: If a disease develops at a certain anatomical site, a *Finding Site* relation exists between the *Disease* entity and the *Body Structure* entity.

Associated Exam: The *Associated Exam* relation exists between a *Disease* entity and a *Laboratory Exam*, a *Physical Exam*, or a *Radiographic Exam* entity, in order to show the medical correlation between them.

Admission: If a disease falls in a department's scope of admission, an *Admission* relation exists between the *Disease* and the *Medical Department* entity.

Part of: If a *Body Structure* entity is considered as a component of another *Body Structure* entity in a clinical context, there exists a *Part of* relationship between them.

Associated Indicator: An *Associated Indicator* relation would exist between a *Disease* entity and a *Biochemical Indicator* entity, showing their correlation.

Measurement Method: A *Measurement Method* relation exists between a *Biochemical Indicator* entity and an exam entity (*Laboratory Exam*, *Physical Exam*, *Radiographic Exam*), meaning that the biochemical indicator belongs to the scope of the test results.

Complication: A *Complication* relation exists between two *Disease* entities to demonstrate that one is a common complicating disease of the other.

Active Ingredient: A *Active Ingredient* relation describes the correlation between a *Medicinal Substance* entity and a *Medicinal Product* entity, while the former is the component or common name of the latter.

Using: If a *Procedure* entity uses a specific *Medicinal Substance* or textitMedicinal Product, then the *Using* relation exists between the two.

Procedure Site: If a *Procedure* entity is performed on a certain *Body Structure* entity, the *Procedure Site* relation would link the two entities together.

Causative Agent: If an *Organism* entity is a pathogen of a *Disease* entity, then the *Causative Agent* relation exists between them.

Treatment: If a drug or a procedure is applicable to a disease, then a *Treatment* relation links them together. Therefore, a *Treatment* relation exists between a *Disease* entity and a *Procedure*, *Therapy and Assessment*, *Medicinal Substance* or a *Medicinal Product* entity.

Anti-inflammatory: If a drug is used for anti-inflammation for a certain disease, an *Anti-inflammatory* relation exists between them.

Anti-viral If a drug is used for anti-virus for a disease, an *Anti-viral* relation links them together.

Antipyretic: If a drug works as an antipyretic for a disease, an *Antipyretic* relation exists between them.

Pain Relief: If a drug copes with pain for a certain disease, a *Pain relief* relation exists between them.

3.2. Data collection and preprocessing

To acquire more diversified and comprehensive medical knowledge, multiple heterogeneous information sources are employed in the construction process. However, different sources vastly differ in text structures and points of attention. Therefore, we standardize the medical texts from various sources for subsequent knowledge extraction. Next, we will introduce the information sources used for the construction of *LMKG*.

First, four wiki platforms are chosen: A-plus Hospital Wiki, China Medical Information Platform, Xunyiwenyao Medical Wiki, and Shiliaotong Food Wiki. A-plus hospital wiki is an open online medical encyclopedia website that can be edited by the public.[2] It contains abundant and useful knowledge of various medical concepts such as diseases, symptoms, and drugs. China Medical Information Platform provides authoritative medical information queries for disease, symptoms, drugs, health care products, and medical apparatus.[3] The Xunyiwenyao Medical Wiki is a medical platform aimed at accurate medical information inquiry, providing both medical knowledge and online consultation for patients.[4] The Shiliaotong Food Wiki is a functional knowledge base of food ingredients, covering nutritional fortification, bacterial strains, medicinal food, health food, and common food ingredients.[5]

To improve the professionalism of the constructed KG, academic medical papers are also used for knowledge extraction. Academic papers are collected from two medical journal databases: the Chinese Medical Journal Network and the Journal of Medical Hepatology Database. Chinese Medical Journal Network is a collection platform for journal papers consisting of 200 journals and over 1,000,000 papers.[6] Journal of Medical Hepatology is a medical journal specializing in hepatobiliary diseases.[7] These two journal databases can provide professional medical knowledge for *LMKG*.

Besides, we also collect data from several official databases providing standard information for medicinal products. The first one is the "National Medical Product Administration Data Search" database,[8] which provides abundant knowledge for the state-approved drug information. The second one is "The State Administration for Market Regulation" database, which provides a catalog of all the marketed drugs.[9] The third one is a professional, large-scale clinical terminology named Snomed CT [42]. It aims to determine international standards for clinical and healthcare terms.[10] These official databases can provide accurate information to enhance the knowledge of *LMKG*.

3.3. Entity recognition

As a wildly studied task, various approaches have been applied to medical NER. Based on the experimental results from Dai et al. [24], the BERT-BiLSTM-CRF model can outperform other architectures on several different datasets. Therefore, we employ a BERT-BiLSTM-CRF-based model architecture in our framework for medical entity recognition on different information sources. Specifically, it is a combination of the BERT embedding layer, the bidirectional LSTM layer, and the CRF layer. The BERT embedding layer can provide both character representations with rich information and the underlying semantic connections between characters [43]. The bidirectional LSTM layer efficiently uses both past and future input features, and the CRF layer aims to identify the optimum chain of labels for a given input sentence by learning the correlations between labels in neighborhoods [16].

Given a sentence $\mathbf{c} = [c_1, c_2, \dots, c_N]$, where c_i denotes the ith token of the sentence, and N denotes the maximum sentence length. We use the Chinese BERT model to map each token c_i to its corresponding character embedding e_i. The last two layers of BERT remain trainable in the training process. Then, the embeddings $\mathbf{e} = [e_1, e_2, \dots, e_N]$ are fed into the BiLSTM layer, which can extract features from both the past and future in the sentence. The hidden states of the forward LSTM $\mathbf{h}_t = [\overrightarrow{h_1}, \overrightarrow{h_2}, \dots, \overrightarrow{h_N}]$ are concatenated together with the hidden states of the backward LSTM $\mathbf{h}_t = [\overleftarrow{h_1}, \overleftarrow{h_2}, \dots, \overleftarrow{h_N}]$. In this way, we have the complete output of BiLSTM $\mathbf{h} = [h_1, h_2, \dots, h_N]$. Next, the Conditional Random Field (CRF) layer is used to model the dependencies between the adjacent tags. Let $\mathbf{y} = [y_1, y_2, \dots, y_N]$ be a sequence of labels for

2 http://www.a-hospital.com.

3 https://www.dayi.org.cn.

4 http://jib.xywy.com.

5 https://slt.21food.cn.

6 http://www.yiigle.com/index.

7 http://www.lcgdbzz.org/.

8 https://www.nmpa.gov.cn/datasearch/home-index.html.

9 https://www.cde.org.cn/hymlj/index.

10 https://www.snomed.org.

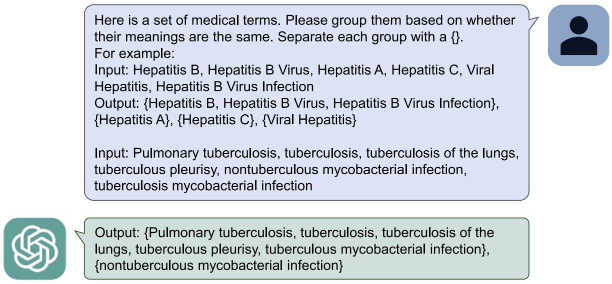

Fig. 4. Example of using chatGPT for medical term grouping for entity alignment. The prompt provides medical terms to be grouped and also the correct format of the expected output.

the sentence $\mathbf{c}$, and $\mathcal{Y}_c$ is the set of all possible $\mathbf{y}$. The score for $\mathbf{y}$ is calculated as:

$$m(\mathbf{h},\mathbf{y}) = \sum_{i=1}^{N} L_{i,y_i} + \sum_{i=1}^{N-1} T_{y_i,y_{i+1}}, \tag{1}$$

$$\mathbf{L}_i = \mathbf{W}_c \mathbf{h}_i + \mathbf{b}_c, \tag{2}$$

where L_{i,y_i} is the local score of the ith token, $T_{y_i,y_{i+1}}$ is the transition score of the two adjacent labels y_i and y_{i+1}. $\mathbf{W}_c$, $\mathbf{b}_c$ and $\mathbf{T}$ are trainable parameters of the CRF layer. Then, the probability of sequence $\mathbf{y}$ being the labels of sentence c can be modeled by the softmax function:

$$p(\mathbf{y}|\mathbf{c}) = \frac{\exp(m(\mathbf{h},\mathbf{y}))}{\sum_{\mathbf{y}'\in\mathcal{Y}_c} m(\mathbf{h},\mathbf{y}')}. \tag{3}$$

Finally, the log-likelihood is used as the loss function:

$$\mathcal{L} = -\sum_{\mathbf{c}^{(n)}} \log p(\mathbf{y}^{(n)}|\mathbf{c}^{(n)}). \tag{4}$$

We use the CCKS-18 dataset [44], a Chinese medical records dataset with annotated medical entities of different types containing 1000 records, to train our medical NER model for KG construction.

3.4. Relation extraction

Based on the experimental results from Qi et al. [31], we employ their medical RE framework and slightly modify it for our task. First, we use Chinese BERT embedding to generate embeddings with rich semantic information. Then the BiLSTM model is utilized for semantic meaning encoding. Let $[\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_N]$ be the matrix of the output of the Bi-LSTM model, where N is the length of the sentence. Next, the attentive pooling method is used to obtain the embedding representation of the head entity, the tail entity, and the sentence itself. Let the positions of the head entity's boundary characters be p^s_{head} and p^e_{head}, we first calculate the attention scores of its characters:

$$a_j = \mathbf{q}^T \cdot \tanh(\mathbf{W}^a \mathbf{w}_j + \mathbf{b}^a), \quad p^s_{head} \le j \le p^e_{head}, \tag{5}$$

where a_j is the attention score of the jth character for learning representation of the head entity, $\mathbf{q}$ is the trainable attention query vector, $\mathbf{W}^a$ and $\mathbf{b}^a$ are trainable parameters of the attention network.

Based on the above, we can obtain the representation of the head entity $\mathbf{e}_{head}$ by attentively aggregating representations of its characters:

$$\mathbf{e}_{head} = \sum_{j=p^s_{head}}^{p^e_{head}} \alpha_j \mathbf{h}_j, \quad \alpha_j = \frac{\exp(a_j)}{\sum_{k=p^s_{head}}^{p^e_{head}} \exp(a_k)}, \tag{6}$$

where α_j is the attention score of the jth character in the head entity. Next, we can obtain the embeddings for the tail entity and the whole sentence based on the same method as the head entity. The three embeddings are denoted as $\mathbf{e}_{head}$, $\mathbf{e}_{tail}$, and $\mathbf{e}_{sen}$, respectively.

Since the types of the head entity and tail entity are known, only a part of medical relations are potential prediction results. Thus, we further encode the head entity type, tail entity type, and potential relation type separately into three embedding representations t_{head}, t_{tail}, and t_{rel}, and convert the task into predicting whether the relationship t_{rel} represents exists between the two entities that t_{head} and t_{tail} represent. Therefore, the three semantic embedding representations e_{head}, e_{tail}, and e_{sen}, along with the three type embeddings t_{head}, t_{tail}, and t_{rel}, are all fed into a binary classifier for relation classification.

The RE model is trained on the CmeIE (Chinese Medical Information Extraction) dataset from the CBLUE (Chinese Biomedical Language Understanding Evaluation) benchmark [45]. The CmeIE dataset has over 20,000 documents, containing nearly 75,000 triples and 53 types of medical relations.

3.5. Hierarchical entity alignment

Inevitably, medical knowledge extracted from different sources has a certain level of redundancy due to duplication of information. Therefore, entity alignment is a necessary step for the refinement of the knowledge graph constructed based on multiple information sources. In order to eliminate redundant entities across different sources, we propose a hierarchical entity alignment framework for the improvement of the extracted knowledge. Generally, entity alignment aims to identify entities across different graphs that refer to the same real-world object. Our proposed framework consists of two procedures: coarse-grained alignment based on similarity and fine-grained refinement using chatGPT.

The main idea of the coarse-grained alignment procedure is to model the relevance between entities at hierarchical views, including the attribute view and the entity view. First, at the attribute view, the Jaccard similarity coefficient is used to define the similarity between the same attributes of different entities. Given an attribute string m and an attribute string n, the attribute similarity coefficient is defined as:

$$J_a = \frac{A \cap B}{A \cup B}, \tag{7}$$

where A and B are the sets of all characters of the strings m and n, respectively.

Next, at the entity view, we calculate the entity similarity coefficient based on the weighted sum of the attribute similarity:

$$J_e = \sum_{a_i \in D} \alpha_i J_a^i, \tag{8}$$

where D is the set of overlapping attributes of the two entities, and J_a^i is the attribute similarity coefficient of the ith attribute in D. The weights α_i are hyperparameters that can be adjusted for each attribute a_i. Based on the entity view similarity coefficient J_e, different entities with high similarity scores are merged as one, along with their attributes and edges. Specifically, we maintain the attribute with a longer value and all the unique attributes while merging entities. We have set a threshold for similarity, where entities with a similarity score above this threshold will be merged together.

After conducting a coarse-grained entity alignment based on similarity, initial knowledge integration from different sources has been achieved. However, this procedure alone may not meet the required level of accuracy to support intelligent medical applications. The alignment process mainly relies on the structural information and semantic textual information of the extracted knowledge. While both types of information heavily rely on the quality of the data source, there is a significant amount of noise in the extracted raw knowledge. Moreover, a considerable number of entities lack textual descriptions, making it difficult to match and integrate them from different sources. In addition, it is challenging to directly align multilingual texts based on text similarity due to variations in information source languages. In order to address these issues, we employ ChatGPT for refining entity alignment.

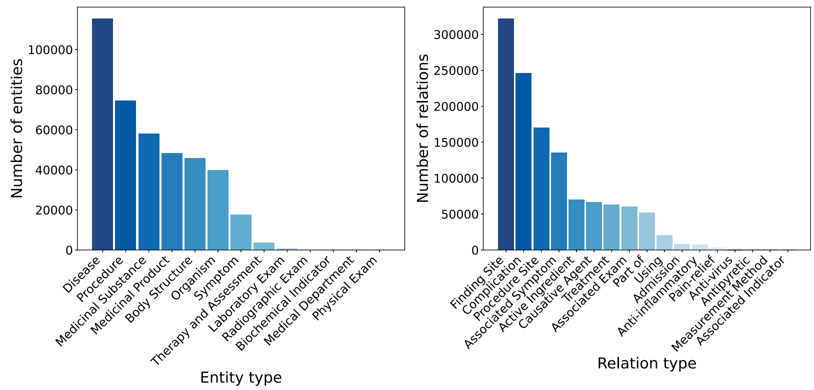

Fig. 5. The entity and relation types distribution as well as the 1-hop, 2-hop neighbor distribution.

ChatGPT has undergone comprehensive pre-training, thereby minimizing the need for additional data to perform diverse downstream tasks. By engaging in dialogue, one can uncover the knowledge retained by ChatGPT from pre-training. Consequently, it can proficiently accomplish bilingual entity alignment by solely requiring entity names. The test results of randomly selected samples have demonstrated ChatGPT's satisfactory refinement effects in the domains of bilingual translation, monolingual, and cross-lingual entity alignment.

In order to enhance the efficiency of utilizing ChatGPT, we aim to minimize the number of questions posed and enhance the uniformity of responses. To achieve these objectives, we have developed specific techniques for prompt engineering. Specifically, we ask for the grouping of a set of terms with low recall thresholds and provide an example as output specification. The lower similarity threshold reduces the chances of missing out on coarse-grained alignments, providing more terms that may refer to the same entity for GPT to filter. The top-down semantic clustering grouping mode effectively reduces the total number of questions asked, saving time and cost. The prompt provides an example with a standardized format, making it easier to parse groups from the answers in the later stage. Fig. 4 shows an example of ChatGPT grouping.

3.6. Knowledge graph and its management system

Based on the procedure introduced above, we extracted a medical KG containing over 403 thousand entities and 1225 thousand relations. The information source contains over 3000 MB of text data. We have released the resources of the constructed KG for download, in the hope of assisting medical research and applications. A query and management interface of *LMKG* has also been provided to facilitate public utilization. Our open-source project is based on the MIT license, which allows users to freely use, modify, and distribute the resources. Detailed instructions on KG data resources and management interface can be found on https://github.com/ypr17.

4. Experiment

4.1. Statistics

We compared the number of entity instances and relation instances, as well as the number of entity types and relation types between *LMKG* and KGs in other works, as shown in Table 1. We also show the entity and relation distribution of *LMKG*, as well as the 1-hop, 2-hop neighbor distribution in Fig. 5. From these results, we find that *LMKG* outperforms existing KGs in different aspects including knowledge scale, the number of entity types and relation types, and the diversity of information sources. In addition to common entity types that most KGs possess, such as *Disease*, *Symptom*, and *Medicinal Product*, etc., rarer and more concrete entity types such as *Organism* , *Biochemical Indicator*, *Radiographic Exam*, *Laboratory Exam*, *Therapy and Assessment* and *Medical Department*, etc. are also covered in *LMKG* to provide a thorough description of real-world medical knowledge. With the fine-grained entity and relation type division, *LMKG* can better facilitate research and applications in different scenarios.

4.2. Knowledge graph embedding

Graphs are composed of nodes and edges, which can be represented by a high-dimensional matrix. However, the graph data can be difficult to interpret by most popular intelligent algorithms, such as machine learning and deep learning methods. To solve this problem, the knowledge graph embedding technique is used to map graph data into real vector space, making it easier for intelligent algorithms to utilize the information buried in graph data.

We employed four different knowledge representation methods, namely TransE [46], DistMult [47], ComplEx [48], and HolE [49], to generate knowledge graph embeddings for *LMKG*. In order to further improve the effectiveness of knowledge representations, we initialize the models with the CLS token embedding of the entity names modeled with BERT. For visualization, the t-SNE [50] method is used to reduce the dimension of the graph embeddings from the original 768-dimension to a 2-dimension planar. After dimension reduction, we can visualize the embeddings for intuitive understanding, as shown in Fig. 6. The visualizations of graph embeddings based on four different knowledge representation methods all clearly show the clustering effect of entities, which demonstrates that the knowledge representations can effectively encode the semantic concepts of different types of medical entities.

4.3. Quality evaluation

Next, we conduct experiments to evaluate the quality of *LMKG* and compare it with other existing medical KGs. To achieve this goal, we employ the medical KGs to guide the deep learning models for two important medical tasks, i.e., medical NER and RE, and compare the qualities of KGs based on the performance improvement brought by them. Besides, we also employ human evaluation for direct quality assessment with the assistance of professional doctors.

4.3.1. Knowledge-enhanced medical NER

We will elaborate on the evaluation of *LMKG* based on the medical NER task. In our experiments, the combination of KGs and the medical NER model is based on the methods proposed by Wang et al. [22]. Specifically, feature vectors are constructed according to the medical knowledge provided by KGs, and are fed into deep neural networks as decision guidance.

Next, we will introduce the medical NER datasets, medical KGs used for comparison, the basic NER models, and the corresponding experimental results.

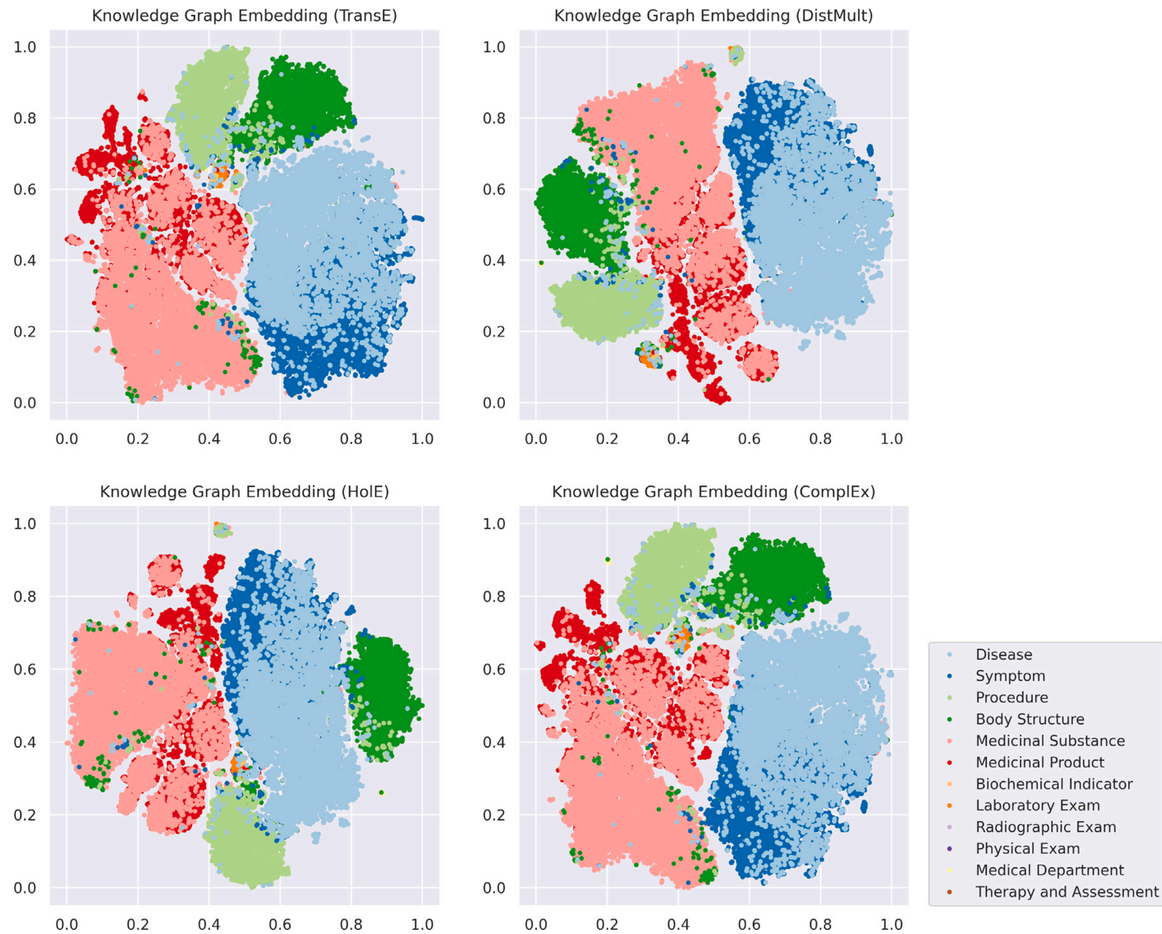

Fig. 6. The visualization of the dimension-reduced knowledge graph embeddings of *LMKG* using four different methods: TransE [46], DistMult [47], ComplEx [48], and HolE [49].

We use two medical data sets for evaluation. The first one is CCKS-19, a medical NER dataset consisting of EMRs covering diseases in general practice [51]. The training set includes 1000 medical records, which are manually labeled with 6 types of entities, including diseases and diagnoses, examinations, tests, procedures, drugs, and anatomies. The test set contains 379 medical records. The second one is Dia-KG, which is a NER and RE data set specialized in diabetes-related medical text materials, containing 22,050 entities and 6890 relations in total, including 18 entity types and 15 relation types [52]. While nested entities are allowed in Dia-KG for disease stages or anatomies, we eliminate them in the evaluation NER task for simplicity.

Two other open-source medical KGs are utilized in our experiments for comparison. The first one is the EMR KG [12], which is a disease–symptom graph extracted from 273,174 patient records, including 647 entities and 3120 relations. The second one is CSKB [36], which is a Chinese symptom knowledge base containing 8722 symptom entities, along with 146 631 relations between the symptoms and related diseases, body parts stroke, and suffering population.

Three medical NER models are implemented respectively: CNN-based [53], BiLSTM-based [54], and BERT-based model [24]. The performance of different models is evaluated by Precision, Recall, and F-1 score. The results are shown in Fig. 7, from which we have several findings. First, all of the used KGs have improved the performance of NER tasks compared with the baseline. This is because all KGs can bring external information into the model. Intuitively, the more precise and comprehensive the KG is, the greater the improvement in the results. That is, the experimental results can reflect the quality of KGs to a certain extent. Second, *LMKG* outperforms other KGs in terms of enhancing the results. This is because existing open-source medical KGs are relatively small in scale with limited knowledge accuracy. Different from these medical KGs, *LMKG* is larger in scale, and more accurate in entity and relation classifications. Therefore, it can enhance the experimental results most effectively. The experimental results can successfully verify the scale, knowledge accuracy, and overall quality of *LMKG*.

4.3.2. Knowledge-enhanced RE

We now introduce the evaluation of *LMKG* based on medical RE. Specifically, we use the graph embeddings to guide the decision-making of the RE model. The graph embedding of head entity e_g^{head}, tail entity e_g^{tail}, and their corresponding relation e_g^{rel} are all incorporated into the model as input features. After obtaining text embeddings of the entities (e_t^{head} and e_t^{tail}) generated with different language models, these three graph embeddings e_g^{head}, e_g^{tail} and e_g^{rel} are concatenated together with the text embeddings of the targeted entities and are fed into the classifier. Again, the Dia-KG data set [52] is used for the RE task. Since other open-source KGs do not provide graph embeddings, we have only evaluated the performance of *LMKG*.

Three commonly used medical RE models are implemented respectively for the evaluation of *LMKG*: CNN-based [55], BiLSTM-based [56], and BERT-based [57] model. We use weighted average Precision, Recall, and F1 score for evaluation. The experimental results are shown in Fig. 8, from which we can see that the RE results are effectively and consistently improved by the graph embeddings of *LMKG*. It is obvious that the three models vary in the capability of language modeling and semantic understanding. Even so, all three models have significantly better results when incorporated with medical knowledge provide by *LMKG*. Therefore, the results can reflect the validity of the graph embeddings and the high quality of *LMKG*.

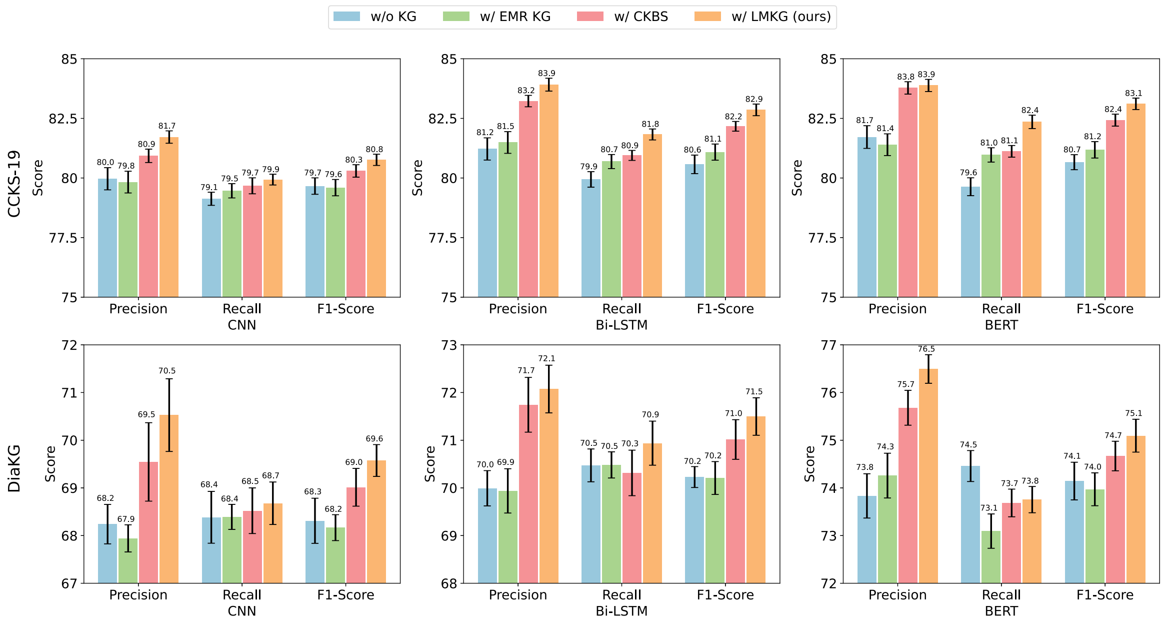

Fig. 7. The results of NER experiments under the guidance of KGs on two real-world biomedical data sets.

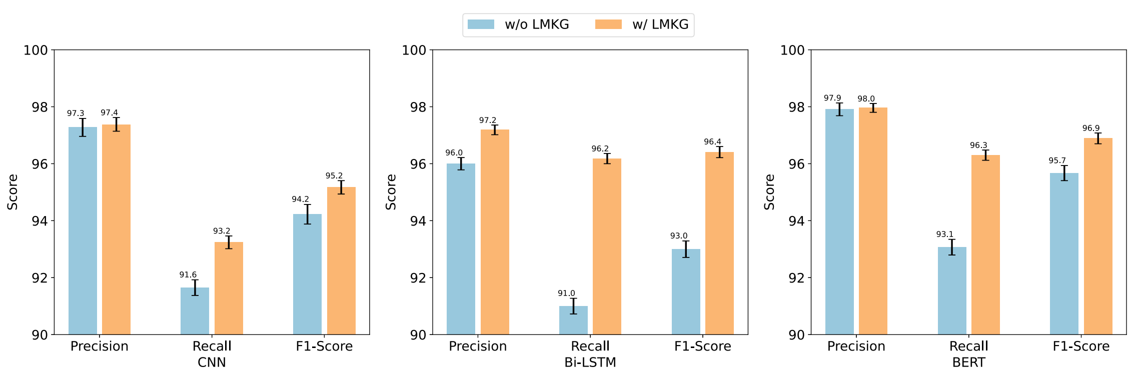

Fig. 8. The results of RE experiments under the guidance of *LMKG* on a real-world biomedical data set.

4.3.3. Manual evaluation

To conduct a more comprehensive evaluation of the quality of *LMKG* from a professional perspective, we sought the assistance of medical experts for manual evaluation. We randomly selected 1000 triplets from *LMKG*, and professional doctors helped assess their correctness. This process allowed us to obtain reliable insights into the correctness and reliability of the triplets. The sampled triplets and corresponding evaluation records can be found at https://github.com/ypr17/LMKG. The results revealed that 94% of the triplets were medically accurate. This manual evaluation confirms that *LMKG* exhibits a high level of quality and is suitable for various intelligent medical tasks.

4.4. Discussions on intelligent applications

The applications of medical knowledge graphs are diverse and can be categorized into two types based on their supporting modules and algorithms. The first type is the applications in upstream tasks, which utilizes the medical KG to support fundamental intelligent medicine algorithms. Examples include KG-enhanced medical NLP tasks such as medical NER and RE, as well as enhancing the performance of medical large language models. In our research, we have evaluated the application in several upstream tasks, and the results are depicted in Figs. 7 and 8. Results indicate that *LMKG* exhibits a high level of quality and can effectively enhance the performance of upstream medical tasks. In future work, we plan to integrate the KG as an auxiliary knowledge base to enhance the performance of medical large language models.

The second type is the applications in downstream tasks, which can be directly utilized by doctors and patients, such as drug discovery and clinical decision support systems (CDSS). In our work, *LMKG* has been applied to a hepatitis B clinical decision support system (CDSS) to provide professional medical knowledge for disease reasoning. In this CDSS, *LMKG* plays a role in assisting with the structuring of medical record text and supporting diagnostic reasoning. Through the entities and relationships *LMKG*, the CDSS can infer the relationship between a patient's symptoms and disease staging, as well as the corresponding treatments.

In order to perform performance testing on the hepatitis B CDSS, 100 outpatient medical records from JD Health are manually annotated by professional doctors, with the annotated labels representing the stages of hepatitis B. According to the evaluation of professional doctors, the CDSS can achieve an accuracy rate of 90%. The interface of the hepatitis B CDSS is on our page: https://github.com/ypr17/HB-CDSS. Overall, results show that *LMKG* can effectively enhance the performance of downstream medical applications.

5. Limitation

In this study, we build a unified system capable of extracting and managing medical knowledge from heterogeneous information sources.

As a result, a high-quality medical KG (named *LMKG*) is constructed to facilitate the development of intelligent applications in the field of medicine. However, there are still some shortcomings and limitations in this work.

Currently, the focus of *LMKG* is mainly on the integration of clinical knowledge, with limited attention given to the domains of pharmacy and biochemistry. Moreover, the relations are now primarily founded on empirical evidence in clinical medicine, lacking the inclusion of logical reasoning and rational inference, such as drug interactions or gene pathogenicity. Therefore, we aim to enhance *LMKG* with supplementary knowledge on pharmacy and biochemistry in future work.

6. Conclusion

In this paper, we construct a high-quality, large-scale, multi-source, and multi-lingual medical KG (named *LMKG*) from heterogeneous medical data. The *LMKG* includes 13 entity types and 17 relation types, and contains 403,784 entity and 1,225,097 relation instances. The scale of *LMKG* is considerably larger than existing medical KGs in both the number of entities and relations and the number of entity types and relation types. We also conduct extensive experiments to verify the superiority of *LMKG*. Experimental results show that *LMKG* can consistently and significantly outperform existing medical KGs in guiding the ML models in downstream medical tasks. We will release *LMKG* and the corresponding graph embeddings to contribute to clinical research and applications.

CRediT authorship contribution statement

Peiru Yang: Writing – original draft, Methodology, Conceptualization. **Hongjun Wang:** Software, Methodology. **Yingzhuo Huang:** Visualization. **Shuai Yang:** Resources, Formal analysis. **Ya Zhang:** Resources. **Liang Huang:** Investigation. **Yuesong Zhang:** Investigation. **Guoxin Wang:** Data curation. **Shizhong Yang:** Validation, Data curation. **Liang He:** Writing – review & editing. **Yongfeng Huang:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

I have shared the link to my data in the manuscript.

Acknowledgments

This work was supported by Key R&D Program in Xinjiang Uygur Autonomous Region, China (2022B03019-6), National Natural Science Foundation of China No. 82090053, and Tsinghua University Initiative Scientific Research Program of Precision Medicine, China under Grant number 2022ZLA007.

References

[1] H. Ren, J. Wang, W.X. Zhao, Generative adversarial networks enhanced pre-training for insufficient electronic health records modeling, in: Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2022, pp. 3810–3818.
[2] X.S. Zhang, F. Tang, H.H. Dodge, J. Zhou, F. Wang, Metapred: Meta-learning for clinical risk prediction with limited patient electronic health records, in: Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining, 2019, pp. 2487–2495.
[3] F. MacLean, Knowledge graphs and their applications in drug discovery, Expert Opin. Drug Discov. 16 (9) (2021) 1057–1069.
[4] X. Zeng, X. Tu, Y. Liu, X. Fu, Y. Su, Toward better drug discovery with knowledge graph, Curr. Opin. Struct. Biol. 72 (2022) 114–126.
[5] S. Sang, Z. Yang, X. Liu, L. Wang, H. Lin, J. Wang, M. Dumontier, GrEDeL: A knowledge graph embedding based method for drug discovery from biomedical literatures, Ieee Access 7 (2018) 8404–8415.
[6] M. Sheng, Q. Hu, Y. Zhang, C. Xing, T. Zhang, A data-intensive CDSS platform based on knowledge graph, in: Health Information Science: 7th International Conference, HIS 2018, Cairns, QLD, Australia, October 5–7, 2018, Proceedings 7, Springer, 2018, pp. 146–155.
[7] X. Xiang, Z. Wang, Y. Jia, B. Fang, Knowledge graph-based clinical decision support system reasoning: a survey, in: 2019 IEEE Fourth International Conference on Data Science in Cyberspace (DSC), IEEE, 2019, pp. 373–380.
[8] G. Zhou, E. Haihong, Z. Kuang, L. Tan, X. Xie, J. Li, H. Luo, Clinical decision support system for hypertension medication based on knowledge graph, Comput. Methods Programs Biomed. 227 (2022) 107220.
[9] H. Saadat, B. Shah, Z. Halim, S. Anwar, Knowledge graph-based convolutional network coupled with sentiment analysis towards enhanced drug recommendation, IEEE/ACM Trans. Comput. Biol. Bioinform. (2022).
[10] X. Wang, Y. Zhang, X. Wang, J. Chen, A knowledge graph enhanced topic modeling approach for herb recommendation, in: Database Systems for Advanced Applications: 24th International Conference, DASFAA 2019, Chiang Mai, Thailand, April 22–25, 2019, Proceedings, Part I 24, Springer, 2019, pp. 709–724.
[11] X. Zou, A survey on application of knowledge graph, in: Journal of Physics: Conference Series, Vol. 1487, IOP Publishing, 2020, 012016.
[12] M. Rotmensch, Y. Halpern, A. Tlimat, S. Horng, D. Sontag, Learning a health knowledge graph from electronic medical records, Sci. Rep. 7 (1) (2017) 1–11.
[13] C. Zhao, J. Jiang, Z. Xu, Y. Guan, A study of EMR-based medical knowledge network and its applications, Comput. Methods Programs Biomed. 143 (2017) 13–23.
[14] L. Li, P. Wang, J. Yan, Y. Wang, S. Li, J. Jiang, Z. Sun, B. Tang, T.-H. Chang, S. Wang, et al., Real-world data medical knowledge graph: construction and applications, Artif. Intell. Med. 103 (2020) 101817.
[15] B. Cheng, J. Zhang, H. Liu, M. Cai, Y. Wang, Research on medical knowledge graph for stroke, J. Healthc. Eng. 2021 (2021).
[16] Z. Huang, W. Xu, K. Yu, Bidirectional LSTM-CRF models for sequence tagging, 2015, arXiv preprint arXiv:1508.01991.
[17] J. Chiu, E. Nichols, Named entity recognition with bidirectional LSTM-CNNs, Comput. Sci. (2015).
[18] Z. Yang, R. Salakhutdinov, W. Cohen, Multi-task cross-lingual sequence tagging from scratch, 2016, arXiv preprint arXiv:1603.06270.
[19] J. Li, A. Sun, J. Han, C. Li, A survey on deep learning for named entity recognition, IEEE Trans. Knowl. Data Eng. 34 (1) (2020) 50–70.
[20] S. Hochreiter, J. Schmidhuber, Long short-term memory, Neural Comput. 9 (8) (1997) 1735–1780.
[21] K. Cho, B. Van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, Y. Bengio, Learning phrase representations using RNN encoder-decoder for statistical machine translation, 2014, arXiv preprint arXiv:1406.1078.
[22] Q. Wang, Y. Zhou, T. Ruan, D. Gao, Y. Xia, P. He, Incorporating dictionaries into deep neural networks for the Chinese clinical named entity recognition, J. Biomed. Inform. 92 (2019) 103133.
[23] J. Li, S. Zhao, J. Yang, Z. Huang, B. Liu, S. Chen, H. Pan, Q. Wang, WCP-RNN: a novel RNN-based approach for Bio-NER in Chinese EMRs, J Supercomput. 76 (3) (2020) 1450–1467.
[24] Z. Dai, X. Wang, P. Ni, Y. Li, G. Li, X. Bai, Named entity recognition using BERT BiLSTM CRF for Chinese electronic health records, in: 2019 12th International Congress on Image and Signal Processing, Biomedical Engineering and Informatics (Cisp-Bmei), IEEE, 2019, pp. 1–5.
[25] J.M. Giorgi, G.D. Bader, Transfer learning for biomedical named entity recognition with neural networks, Bioinformatics 34 (23) (2018) 4087–4094.
[26] M. Miwa, M. Bansal, End-to-end relation extraction using lstms on sequences and tree structures, 2016, arXiv preprint arXiv:1601.00770.
[27] Z. Guo, Y. Zhang, W. Lu, Attention guided graph convolutional networks for relation extraction, 2019, arXiv preprint arXiv:1906.07510.
[28] T.H. Nguyen, R. Grishman, Relation extraction: Perspective from convolutional neural networks, in: Proceedings of the 1st Workshop on Vector Space Modeling for Natural Language Processing, 2015, pp. 39–48.
[29] T. Zhang, H. Lin, M.M. Tadesse, Y. Ren, X. Duan, B. Xu, Chinese medical relation extraction based on multi-hop self-attention mechanism, Int. J. Mach. Learn. Cybern. 12 (2) (2021) 355–363.
[30] C. Ruan, Y. Wu, G.S. Luo, Y. Yang, P. Ma, Relation extraction for Chinese clinical records using multi-view graph learning, IEEE Access 8 (2020) 215613–215622.
[31] T. Qi, S. Qiu, X. Shen, H. Chen, S. Yang, H. Wen, Y. Zhang, Y. Wu, Y. Huang, KeMRE: Knowledge-enhanced medical relation extraction for Chinese medicine instructions, J. Biomed. Inform. 120 (2021) 103834.
[32] Z. Zhang, T. Zhou, Y. Zhang, Y. Pang, Attention-based deep residual learning network for entity relation extraction in Chinese EMRs, BMC Med. Inform. Decis. Mak. 19 (2) (2019) 171–177.
[33] S.G. Finlayson, P. LePendu, N.H. Shah, Building the graph of medicine from millions of clinical narratives, Sci. Data 1 (1) (2014) 1–9.

[34] X. Zhou, J. Menche, A.-L. Barabási, A. Sharma, Human symptoms–disease network, Nat. Commun. 5 (1) (2014) 4212.
[35] H. Weng, J. Chen, A. Ou, Y. Lao, et al., Leveraging representation learning for the construction and application of a knowledge graph for traditional Chinese medicine: Framework development study, JMIR Med. Inform. 10 (9) (2022) e38414.
[36] H. Zan, Y. Han, Y. Fan, C. Niu, K. Zhang, Z. Sui, 中文症状知识库的建立与分析, 中文信息学报, 4, 2020, p. 8.
[37] P. Ernst, A. Siu, G. Weikum, Knowlife: a versatile approach for constructing a large knowledge graph for biomedical sciences, BMC Bioinform. 16 (2015) 1–13.
[38] R. Celebi, H. Uyar, E. Yasar, O. Gumus, O. Dikenelli, M. Dumontier, Evaluation of knowledge graph embedding approaches for drug-drug interaction prediction in realistic settings, BMC Bioinform. 20 (1) (2019) 1–14.
[39] L. Li, P. Wang, Y. Wang, S. Wang, J. Yan, J. Jiang, B. Tang, C. Wang, Y. Liu, et al., A method to learn embedding of a probabilistic medical knowledge graph: algorithm development, JMIR Med. Inform. 8 (5) (2020) e17645.
[40] F. Gong, M. Wang, H. Wang, S. Wang, M. Liu, SMR: medical knowledge graph embedding for safe medicine recommendation, Big Data Res. 23 (2021) 100174.
[41] Y. Shen, L. Zhang, J. Zhang, M. Yang, B. Tang, Y. Li, K. Lei, CBN: Constructing a clinical Bayesian network based on data from the electronic medical record, J. Biomed. Inform. 88 (2018) 1–10.
[42] K. Donnelly, et al., SNOMED-CT: The advanced terminology and coding system for eHealth, Stud. Health Technol. Inform. 121 (2006) 279.
[43] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, Bert: Pre-training of deep bidirectional transformers for language understanding, 2018, arXiv preprint arXiv:1810.04805.
[44] China Conference on Knowledge Graph and Semantic Computing 2018, website, 2018, www.ccks2018.cn.
[45] N. Zhang, M. Chen, Z. Bi, X. Liang, L. Li, X. Shang, K. Yin, C. Tan, J. Xu, F. Huang, et al., Cblue: A chinese biomedical language understanding evaluation benchmark, 2021, arXiv preprint arXiv:2106.08087.
[46] A. Bordes, N. Usunier, A. Garcia-Duran, J. Weston, O. Yakhnenko, Translating embeddings for modeling multi-relational data, Adv. Neural Inf. Process. Syst. 26 (2013).
[47] B. Yang, W.-t. Yih, X. He, J. Gao, L. Deng, Embedding entities and relations for learning and inference in knowledge bases, 2014, arXiv preprint arXiv:1412.6575.
[48] T. Trouillon, J. Welbl, S. Riedel, É. Gaussier, G. Bouchard, Complex embeddings for simple link prediction, in: International Conference on Machine Learning, PMLR, 2016, pp. 2071–2080.
[49] M. Nickel, L. Rosasco, T. Poggio, Holographic embeddings of knowledge graphs, in: Proceedings of the Thirtieth AAAI Conference on Artificial Intelligence, AAAI '16, AAAI Press, 2016, pp. 1955–1961.
[50] L. Van der Maaten, G. Hinton, Visualizing data using t-SNE, J. Mach. Learn. Res. 9 (11) (2008).
[51] China Conference on Knowledge Graph and Semantic Computing 2019, website, 2019, www.ccks2019.cn.
[52] D. Chang, M. Chen, C. Liu, L. Liu, D. Li, W. Li, F. Kong, B. Liu, X. Luo, J. Qi, et al., DiaKG: An annotated diabetes dataset for medical knowledge graph construction, in: China Conference on Knowledge Graph and Semantic Computing, Springer, 2021, pp. 308–314.
[53] X. Dong, L. Qian, Y. Guan, L. Huang, Q. Yu, J. Yang, A multiclass classification method based on deep learning for named entity recognition in electronic medical records, in: 2016 New York Scientific Data Summit (NYSDS), IEEE, 2016, pp. 1–10.
[54] B. Ji, R. Liu, S. Li, J. Yu, Q. Wu, Y. Tan, J. Wu, A hybrid approach for named entity recognition in Chinese electronic medical record, BMC Med. Inform. Decis. Mak. 19 (2) (2019) 149–158.
[55] S.K. Sahu, A. Anand, K. Oruganty, M. Gattu, Relation extraction from clinical texts using domain invariant convolutional neural network, 2016, arXiv preprint arXiv:1606.09370.
[56] T. Chen, Y. Hu, Entity relation extraction from electronic medical records based on improved annotation rules and BiLSTM-CRF, Ann. Transl. Med. 9 (18) (2021).
[57] C. Lin, T. Miller, D. Dligach, S. Bethard, G. Savova, A BERT-based universal model for both within-and cross-sentence clinical temporal relation extraction, in: Proceedings of the 2nd Clinical Natural Language Processing Workshop, 2019, pp. 65–71.